A Drug Recommendation Model Based on Message Propagation and DDI Gating Mechanism

Yongjian Ren, Yuliang Shi, Kun Zhang, Xinjun Wang, Zhiyong Chen, and Hui Li

***Abstract*—Drug recommendation task based on the deep learning model has been widely studied and applied in the health care field in recent years. However, the accuracy of drug recommendation models still needs to be improved. In addition, the existing recommendation models either give only one recommendation (however, there may be a variety of drug combination options in practice) or can not provide the confidence level of the recommended result. To fill these gaps, a Drug Recommendation model based on Message Propagation neural network (denoted as DRMP) is proposed in this paper. Then, the Drug-Drug Interaction (DDI) knowledge is introduced into the proposed model to reduce the DDI rate in recommended drugs. Finally, the proposed model is extended to Bayesian Neural Network (BNN) to realize multiple recommendations and give the confidence of each recommendation result, so as to provide richer information to help doctors make decisions. Experimental results on public data sets show that the proposed model is superior to the best existing models.**

***Index Terms*—Health informatics, drug recommendation, deep learning, data mining, Bayesian neural network.**

I. INTRODUCTION

THE rapid accumulation of Electronic Health Records (EHRs) provides powerful support for researchers in the field of medical health and deep learning [1]–[4]. The drug recommendation task [5]–[8] recommends reasonable drug combinations for patients according to their health status. It can provide decision support for doctors and has a good prospect in the field of health. Drug recommendation models based on deep learning can map medical entities into implicit vector representations and mine the association between drugs and patients' health status from a wide range of case data.

The existing drug recommendation methods can be divided into case-based recommendation methods and longitudinal recommendation methods. The case-based approach focuses on the patient's current health status. For example, LEAP [9] extracts feature information from current encounterer for drug recommendations. The longitudinal method takes advantage of the time dependence of visit records. For example, RETAIN [10], GAMENet [11], SafeDrug [5], and SARMR [12] deploy sequential neural networks (such as RNN and GRU [13]) to capture the sequential information of visit records. Recently, researchers have tried to reduce the probability of adverse DDI in recommended results. Therefore, GAMENet [11] and SafeDrug [5] introduced additional DDI knowledge, while SARMR [12] proposed a method called self-supervised adversarial regulation. The DDI loss function proposed by SafeDrug [5] can directly act on the loss function and obtain flexible DDI regulation ability. In addition, in health care research, many neural network structures (such as Dipole [14], RETAIN [10], HiTANet [15], and RAIM [16]) adopt attention mechanisms. The attention mechanism acting on different time steps makes the state of each time step propagate forward faster. Thus, the models converge faster and obtain better prediction ability. However, they still suffer from the following limitations.

1) The accuracy of drug recommendation still needs to be improved, which directly affects the deployment prospect and the willingness of doctors.
2) Lacking interpretability of DDI intensity. Although some existing models introduce DDI knowledge to reduce the DDI rate of recommendation results, existing models can't tell us which interactions are milder and which are more severe.
   Therefore, it is of great practical significance to give DDI intensities of recommendation results, which can help doctors make decisions.
3) Lacking diversity and confidence in recommendation results. In practice, there may be a variety of drug combinations that are appropriate for the patient's health. However, most of the existing drug recommendation models only provide one drug combination. Although a few existing models can give multiple drug combinations, they still lack the confidence evaluation for different drug combinations.

Manuscript received July 31, 2021; revised November 8, 2021 and January 9, 2022; accepted February 15, 2022. Date of publication February 23, 2022; date of current version July 4, 2022. This work was supported by the National Key Research and Development Plan of China under Grant 2018YFC0114709. *(Corresponding author: Yuliang Shi.)*

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Digital Object Identifier 10.1109/JBHI.2022.3153342

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In order to solve the above challenges, DRMP is proposed is this paper. The contributions are as follows:

- The DRMP model is proposed. The patient representation module of the proposed model includes three submodules. (a) Taking the temporal medical records as a temporal graph, the message propagation neural network can not only realize the propagation of temporal information but also pay attention to the important temporal association. (b) A skip GRU module is deployed to obtain visit representations. (c) The deployment of visit-level attention enables the model to focus on important visits.
- A gating mechanism for DDI is proposed. With only a few parameters, the model can limit the frequency of DDI in recommendation results. In addition, the DDI intensity can be easily obtained via the weight parameter in DDI gating mechanism.
- The decoding module of the model is extended to BNN, so that it can recommend multiple drug combinations equipped with the confidence level.

II. RELATED WORK

A. Drug Recommendation

RETAIN [10] proposed a neural network model based on a two-level attention mechanism to capture important visits and important features, and provide interpretability for recommendation results while improving modeling performance. Subsequently, DMNC [17] used memory-enhanced neural networks to simulate the interaction between asynchronous two view sequences in EHRs in drug prescription tasks and disease progression tasks. G-BERT [18] learns the hierarchical knowledge of drugs and diagnosis based on GNN, and learns the visit-level representation based on Transformer. The model uses patients with a single visit for pre-training, and then fine-tune the pretrained visit encoder and representations for the prediction task of patients with multiple visits. However, these methods do not consider the relationship between drugs, so GAMENet [11] establishes a graph based on drug co-occurrence, and uses GNN to obtain the embedding of the drug. LEAP [9] makes drug recommendations from the perspective of sequential decision-making. While using the cyclic decoder to model drug dependence, external clinical knowledge is incorporated into the design of enhanced reward to effectively prevent adverse drug combinations. SafeDrug [5] deployed a message propagation network and a bipartite graph learning module to capture the structural information of drug molecules. At the same time, a controllable loss function was deployed to effectively control the DDI level in the recommended drug combination. SARMR [12] obtains the target distribution associated with safe medication combinations from raw patient records for adversarial regularization. In this way, the model can shape distributions of patient representations to achieve DDI reduction. However, these methods rarely consider the diversity and credibility of recommendation results.

B. Interpretability and Trustworthiness

In clinical practice, the interpretability of the model requires that the model can identify the contribution of each medical event to the corresponding prediction results. Recently, various interpretable prediction models have been proposed to facilitate decision-making, such as RETAIN [10] and Dipole [14]. RETAIN describes interpretability, which uses a two-level attention mechanism to detect influential past visits and important clinical variables in these visits. Then, based on the work of RETAIN, INPREM [19] extends the model to BNN to provide uncertainty estimation, so as to evaluate the trustworthiness of the prediction results.

III. PROBLEM DESCRIPTION

A. Patient Data

Let $\{x_1, x_2, \ldots, x_t, \ldots, x_L\}$ represent a medical visit sequence. x_t represents the t^{th} visit, and L represents the length of the sequence. Let D, P, and M denote the set of diagnoses, the set of procedures, and the set of drugs, respectively. $N = |D| + |P| + |M|$ represents the total number of medical items, where $|D|$ represents the size of the set D. The diagnoses, procedures, and drugs are numbered uniformly, and the number range is $0 \sim N-1$. x_t is a list. If a medical item appears in the t^{th} visit, the number of the medical item exists in x_t. The length of x_t is denoted as N_t, which means that N_t medical items appeared in the t^{th} visit.

B. Drug Recommendation Task

Given a sample $s = (x, y)$, the goal of the medication prediction task is to predict the drug set y of the current visit based on the patient's known medical visit information x. $x = \{x_1, x_2, \ldots, x_{L-1}, x'_L\}$, where x'_L represents the last visit in which the diagnoses and procedures are known, but the drug set is not known. $y \in \{0,1\}^{1\times|M|}$ is a vector in which the value range of each element is $\{0,1\}$, where $|M|$ represents the size of drug set M. A value of 1 indicates that the corresponding drug appears, and a value of 0 indicates that the corresponding drug does not appear.

IV. METHOD

As shown in Fig. 1, the model includes a patient representation module, a decoding module, and a DDI gating module.

A. Patient Representation Module

1) Embedding of Medical Items: Let $E \in \mathbb{R}^{N\times dim}$ denote the embedding representation of medical items, where dim denotes the embedding dimension. Given a sample $s = (x, y)$, $x_{t,i}$ represents the number of the i^{th} medical item of the t^{th} visit. Then $e_{t,i}$ represents the embedded representation of the i^{th} medical item that appears in the t^{th} visit. $e_{t,i} \in \mathbb{R}^{1\times dim}$. $e_t \in \mathbb{R}^{N_t\times dim}$ represents all medical items that appeared in the t^{th} visit, where N_t is the number of medical items appeared.

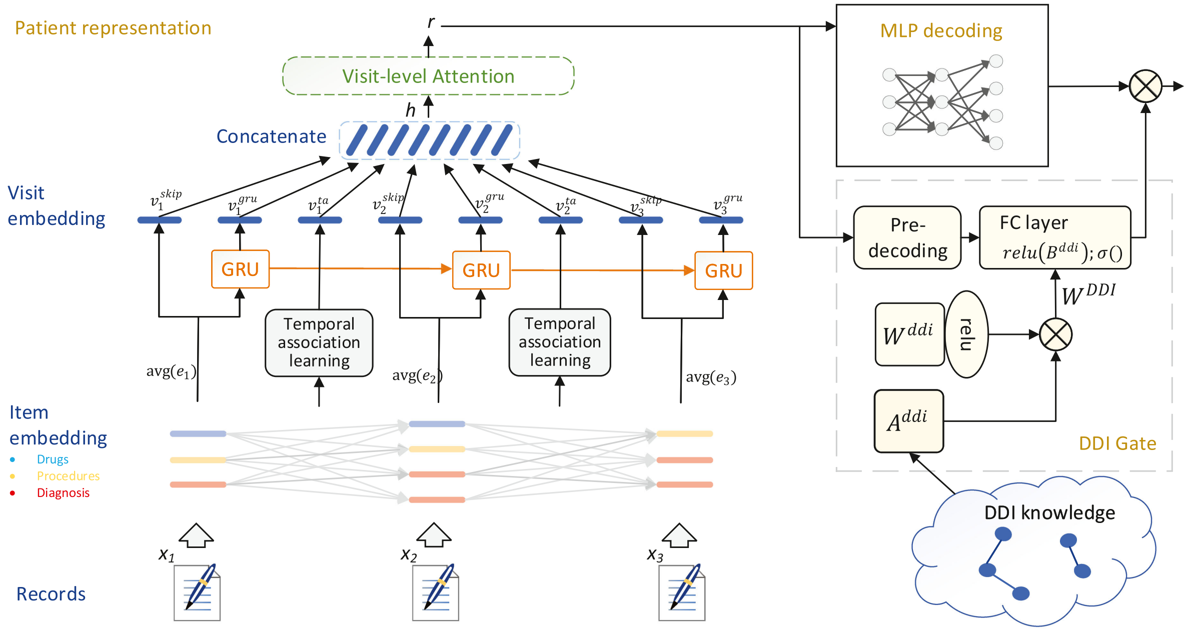

Patient representation
r
MLP decoding
$\hat{y}$
Visit-level Attention
h
Concatenate
Visit embedding
v_1^{skip}
v_1^{gru}
v_1^{ta}
v_2^{skip}
v_2^{gru}
v_2^{ta}
v_3^{skip}
v_3^{gru}
GRU
GRU
GRU
Temporal association learning
Temporal association learning
avg(e_1)
avg(e_2)
avg(e_3)
Pre-decoding
FC layer $relu(B^{ddi}); \sigma()$
W^{DDI}
W^{ddi}
relu
A^{ddi}
DDI Gate
Item embedding
Drugs
Procedures
Diagnosis
x_1
x_2
x_3
Records
DDI knowledge

Fig. 1. Drug recommendation model based on message propagation network and DDI gating mechanism. (1) The patient presentation module contains three sub-modules. (a) Temporary association learning. The patient's visit records are regarded as a temporal graph. KGAT [20] which is a message propagation network is used to learn message propagation in the temporal graph. (b) Skip GRU is used to learn visits' representation. (c) Visit-level attention is used to obtain the patient representation. (2) The DDI gating module generates a gate vector to control the final output of the model. To obtain the interpretable DDI intensity, the weight parameter is set to not less than 0. (3) An MLP decoding module is used to predict drug combinations. The decoding module is extended to BNN, so that the model can output a variety of drug combinations and give the corresponding confidence.

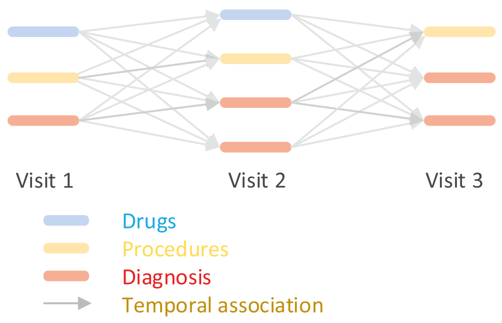

Visit 1
Visit 2
Visit 3
Drugs
Procedures
Diagnosis
Temporal association

Fig. 2. Medical temporal graph. Each medical item in the medical visit is regarded as a node in the temporal graph. The relationship "temporal association" is built between two medical items from adjacent visits. The importance of "temporal association" depends on the learning of the message propagation neural network (e.g., KGAT [20]).

2) Temporal Association Learning: The patient's medical records are treated as a temporal graph, as shown in Fig. 2. Given two adjacent visits x_{t-1} and x_t, each medical item in x_{t-1} and x_t is regarded as a node in the graph. Let a denote a medical item in x_t and b denote a medical item in x_{t-1}, then an edge between b and a is built. This edge represents a relationship, named "temporal association". This relationship is used to describe the possible temporal correlation between the occurrence of one medical item and the occurrence of another medical item.

KGAT [20], a neural network for message propagation, is deployed to capture the importance of the "temporal association" relationship. A medical item $e_{t,i}$ is regarded as a node in the medical graph, and each medical item in e_{t-1} is regarded as a neighbor of $e_{t,i}$. The node aggregation operation of KGAT is used to realize the spread of information from e_{t-1} to $e_{t,i}$.

$$p_{t,i} = \begin{cases} aggr(e_t, e_{t+1,i}), ift = 1 \\ aggr(p_{t-1}, e_{t+1,i}), ift > 1 \end{cases} \tag{1}$$

where $p_{t,i} \in \mathbb{R}^{1\times dim}$ represents a temporal association. The message propagation method $aggr(\cdot)$ is described in detail at the end of this part. Obviously, when $t > 1$, the information of $p_{t,i}$ not only comes from the $t-1^{th}$ visit, but from all visits before the t^{th} visit. The attention mechanism in KGAT can judge the importance of different neighbors, so as to realize the capture of important "temporal association" relationships.

KGAT realizes the update of node representation by aggregating neighbor nodes. In this paper, since the updated representation captures the "temporal association" relationship of medical items, the updated node representation is regarded as a temporal association. Obviously, there are N_t medical items in the visit $t(t \geq 1)$, therefore KGAT will generate N_t temporal associations. The i^{th} temporal association is denoted as $p_{t,i}$.

Thus, the visit-level representation of the temporal association (denoted as v_t^{ta}, where ta is the abbreviation of **t**emporal **a**ssociation) can be obtained. $v_t^{ta} \in \mathbb{R}^{1\times dim}$.

$$v_t^{ta} = avg(p_t) = \frac{1}{N_t}\sum_{1}^{N_t} p_{t,i} \tag{2}$$

where p_t is the matrix consist of $p_{t,i}$.

3) Skip GRU Module: First, a GRU [13] is used to obtain a visit-level representation (denoted as v_t^{gru}).

$$v_t^{gru} = \mathrm{GRU}(avg(e_t)), v_t^{gru} \in \mathbb{R}^{1\times dim} \tag{3}$$

where $avg(e_t) = \frac{1}{N_t}\sum_1^{N_t} e_{t,i}$. Then, the skip connection is deployed to speed up the convergence of the model.

$$v_t^{skip} = avg(e_t), v_t^{skip} \in \mathbb{R}^{1\times dim} \tag{4}$$

where v_t^{skip} represents the visit-level representation obtained via skip connection.

4) Patient's Presentation Learning: The visit-level attention is used to obtain patient representation. $h = [v^{ta}; v^{gru}; v^{skip}]$ is the concatenation of v^{ta}, v^{gru} and v^{skip} in the 0^{th} dimension. $v^{ta} \in \mathbb{R}^{(L-1)\times dim}$, $v^{gru} \in \mathbb{R}^{L\times dim}$ and $v^{skip} \in \mathbb{R}^{L\times dim}$.

$$h = [v^{ta}; v^{gru}; v^{skip}] \in \mathbb{R}^{(3L-1)\times dim} \tag{5}$$

$$r = \sum_1^{3L-1} \alpha_i h_i \tag{6}$$

$$\alpha = softmax(tanh(hW_{\alpha 1})W_{\alpha 2}) \tag{7}$$

where $W_{\alpha 1} \in \mathbb{R}^{dim\times dim}, W_{\alpha 2} \in \mathbb{R}^{dim\times 1}$ are attention weight parameters.

Then, the decoder preliminarily recommends the drug combination according to the patient state:

$$\hat{y}' = sigmoid(\mathrm{MLP}(r)) \tag{8}$$

where MLP represents a Multi-Layer Perceptron.

5) Message Propagation Method $aggr(\cdot)$: KGAT [20] has 3 aggregation methods, which are respectively denoted as KGAT-GCN, KGAT-GraphSage, and KGAT-Bi.

Take $p_{t,i} = aggr(e_t, e_{t+1,i}), t = 1$ as an example:

First, generate the attention of the node $e_{t+1,i}$ to the neighbor $e_{t,j}$, and obtain the aggregated neighbor information (denoted as $message$) based on the attention mechanism.

$$score_j = tanh(tanh([e_{t,j}, e_{t+1,i}]W_{\gamma 1})W_{\gamma 2}) \tag{9}$$

where $[e_{t,j}, e_{t+1,i}] \in \mathbb{R}^{1\times 2dim}$ represents the splicing in the feature dimension.

$$\gamma = softmax(score) \tag{10}$$

$$message = \sum_{j=1}^{N_t} \gamma_j e_{t,j} \tag{11}$$

Next, we need to merge $message \in \mathbb{R}^{1\times dim}$ with $e_{t+1,i} \in \mathbb{R}^{1\times dim}$. The difference between the three KGAT aggregation methods is the difference in the merge method. Let FC represents the fully connected layer of the neural network.

- KGAT-GCN:

$$p_{t,i} = \mathrm{FC}(message + e_{t+1,i}) \tag{12}$$

- KGAT-GraphSage:

$$p_{t,i} = \mathrm{FC}([message, e_{t+1,i}]) \tag{13}$$

where $[message, e_{t+1,i}] \in \mathbb{R}^{1\times 2dim}$ represents the concatenation in the feature dimension.

- KGAT-Bi:

$$p_{t,i} = FC_1(message + e_{t+1,i}) + FC_2(message \odot e_{t+1,i}) \tag{14}$$

where $\odot$ represents the elements-wise multiplication.

- KGAT-GRU: In addition, an merge method based on GRU unit is proposed in this paper, which is denoted as KGAT-GRU.

$$r = sigmoid([message, e_{t+1,i}]W_r) \tag{15}$$

$$z = sigmoid([message, e_{t+1,i}]W_z) \tag{16}$$

$$f = tanh([r \odot message, e_{t+1,i}]W_f) \tag{17}$$

$$p_{t,i} = (1 - z)message + z \odot f \tag{18}$$

where W_r, W_z, W_f are weight parameters similar to GRU unit.

When $t > 1$, the propagation method is the same as the above steps, just replace e_t with p_{t-1}.

B. DDI Gating

1) Design of Gating Parameters: After introducing DDI knowledge, the drug interaction matrix A^{ddi} can be obtained.

$$A_{i,j}^{ddi} = \begin{cases} 1, \text{if drug i,j exist interaction} \\ 0, \text{else} \end{cases} \tag{19}$$

In order to quantify the interaction intensity between any two drugs, the parameter matrix $W^{DDI} \in \mathbb{R}^{|M|\times|M|}$ is introduced. Let parameter $W_{i,j}^{DDI}$ denote the inhibition of drug j after the appearance of drug i. We hope that

$$W_{i,j}^{DDI} \begin{cases} \geqslant 0, if\, A_{i,j}^{ddi} = 1 \\ = 0, else \end{cases} \tag{20}$$

For this purpose, we deploy the activation function $relu$ and use the DDI interaction matrix A^{ddi} as the mask.

$$W^{DDI} = relu(W^{ddi}) \cdot A^{ddi} \tag{21}$$

where W^{ddi} is the parameter matrix of the gating mechanism. Symbol $\cdot$ is the element-wise multiplication.

2) Design of Gating Function: Let DDI gating vector $g = G(\hat{y}', W^{DDI})$, where G is the mapping function. We hope that:

i) W^{DDI} can directly reflect the intensity of DDI. For this reason, we need a negative correlation between g and W^{DDI}, that is, the intenser DDI, the smaller g.

ii) The range of g_i is [0,1].

Therefore, we design the mapping function G as:

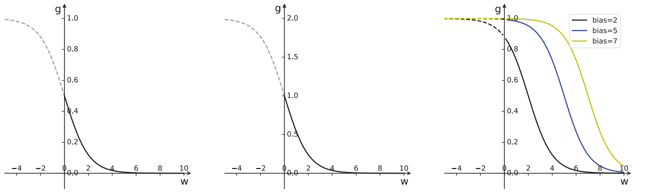
$$g = G_1(W^{DDI}) = 1 - \sigma(\hat{y}' W^{DDI}) \tag{22}$$

where $\sigma(\cdot)$ denotes sigmoid function:

$$\sigma(x) = 1/(1 + e^{-x}) \tag{23}$$

Without losing generality, let $\hat{y}' = 1$. The function image is shown in Fig. 3(a).

In order to satisfy (ii), let $g = 2G_1(W^{DDI})$. The function image is shown in Fig. 3(b). Via the function image, we can see



(a) (b) (c)

Fig. 3. Function study for DDI gating mechanism. Since $W_{i,j}^{DDI} \geqslant 0$, the dotted line in the graph is the point that the function cannot get. Function images with different bias values is shown in subgraph (c). It can be seen that the value of g is more flexible and the expression ability of the model is stronger. (a) $g = 1 - \sigma(W^{DDI})$. (b) $g = 2(1 - \sigma(W^{DDI}))$. (c) $g = 1 - \sigma(W^{DDI} - B^{DDI})$.

that when W^{DDI} changes around 0, it will sensitively affect g, which is not conducive to the convergence and output of the model. To this end, we choose another option to move function G_1 to the right, as shown in Fig. 3(c). Finally, the DDI gating function is obtained:

$$g = G(\hat{y}', W^{DDI}) = 1 - sigmoid(\hat{y}'W^{DDI} - B^{DDI})$$

$$B^{DDI} = relu(B^{ddi}) \geqslant 0 \qquad (24)$$

where $B^{ddi} \in \mathbb{R}^{1\times|M|}$ is the bias vector in the DDI gating mechanism.

The final recommendation result (denoted as $\hat{y}$) of the model is obtained via adjusting the decoding result $\hat{y}'$ by g:

$$\hat{y} = \hat{y}' \cdot g \qquad (25)$$

where symbol $\cdot$ is the element-wise multiplication.

C. Extend Decoding Module to BNN

BNN introduces Bayesian estimation into neural networks to capture and express model uncertainty. Let f denote the neural network, $p(f)$ denote the prior distribution of f. BNN hopes to obtain the posterior distribution $p(f|D_X, D_Y)$ after a given data set (D_X, D_Y). In practice, it is very difficult to use neural networks to calculate posterior distribution. Therefore, researchers resort to approximation to solve this problem [21]. Given a new data point x^*, the prediction can be obtained by integrating the parameter w.

$$p(x^*|y^*, D_X, D_Y) = \int p(y^*|f^*)p(f^*|x^*, w)\, p(w|D_X, D_Y)df^*dw \qquad (26)$$

However, $p(w|D_X, D_Y)$ is also intractable. In this case, the variational inference is proposed, that is, the distribution $q(W)$ is used to approximate the real posterior $p(w|D_X, D_Y)$. By minimizing the KL (Kullback Leibler) divergence of the two distributions, make $q(W)$ be close to $p(w|D_X, D_Y)$.

Applying the Bernoulli distribution to the weight parameters of the neural network, the approximate value of the predicted posterior distribution can be obtained. Therefore, Monte Carlo dropout is used to generate approximate integrals of model weights. Dropout is still active when testing. Given the test x^*, every prediction through the network will provide samples of the posterior distribution. The approximate posterior distribution can be obtained by collecting enough samples.

D. Objective Function

Drug recommendation is a multi-label prediction task. The frequency of different labels may vary greatly. To make the prediction results easier to be classified as 1 and 0, following works [11] and [5], the multi-label-margin loss (denoted as L_{mlm}) is used in addition to binary-cross-entropy loss (denoted as L_{bce}).

$$L_{bce}(\hat{y}, y) = -\sum_i y_i log\hat{y}_i + (1 - y_i)log(1 - \hat{y}_i)$$

$$L_{mlm}(\hat{y}, \iota) = \sum_{i \notin \iota} \sum_{j \in \iota} \frac{max(0, 1 - (\hat{y}_j - \hat{y}_i))}{|M|} \qquad (27)$$

where $y \in \{0,1\}^{|M|}$ is the true label and $\hat{y} \in \mathbb{R}^{|M|}$ is the model prediction result. $\iota = \{i|\forall y_i = 1\}$ represents the set of indexes with positive labels. y_i is the i-th label of y.

In this paper, DDI loss (denoted as L_{ddi}) is defined as:

$$L_{ddi} = \sum_{\forall \bar{y}_i = 1} \sum_{\forall \bar{y}_j = 1} \hat{y}_i * \hat{y}_j * A_{i,j}^{ddi} \qquad (28)$$

where $\bar{y} \in \{0,1\}^{|M|}$ is the recommended labels, i.e. $\bar{y}_i = \begin{cases} 1, if \hat{y}_i \geqslant 0.5 \\ 0, else \end{cases}$.

The total loss (denoted as $Loss$) is:

$$Loss = L_{bce} + \gamma_{mlm} * L_{mlm} + \gamma_{ddi} * L_{ddi} \qquad (29)$$

where γ_{mlm} and γ_{ddi} are hyper parameters used to adjust the 3 losses.

V. EXPERIMENTS

A. Data Set

Experiments on EHRs data based on public data set MIMIC-III [22] are shown in this section. The diagnosis and procedure codes are based on the ICD-9 code. In this paper, the data processing of MIMIC-III and the processing of DDI knowledge are consistent with the processing method in SafeDrug. Specifically, the drug codes in NDC (National Destination Code) in MIMIC-III are transformed into the drug codes in level-3 ATC (Anatomical Therapeutic Chemical) code. The DDI knowledge used includes 40 harmful DDI relationships. DDI knowledge is applied to DRMP in the form of binary knowledge (that is, whether the two drugs have an adverse interaction). For the detailed introduction and acquisition of the NDC-ATC comparison table and DDI knowledge, please refer to SafeDrug's paper [5].

The detailed statistics of the 2 constructed data sets are shown in Table I, where # means "number". In this paper, training set: verification set: test set = 7:1.5:1.5.

B. Baselines

Baselines are divided into two categories. MLP, GRU, RETAIN [10] and SARMR [12] did not introduce additional DDI knowledge, while LEAP [9], GAMENet [11], and SafeDrug [5]

TABLE I
DATA STATISTICS

Statistics	Size
# of patients/visits	6350 / 14995
# of diag./prod./drug	1958 / 1430 / 131
avg./max # of visits	2.37 / 29
avg./max # of diagnoses per visit	10.51 / 128
avg./max # of procedures per visit	3.84 / 50
avg./max # of drugs per visit	11.44 / 65
# of DDI pairs	448

TABLE II
MODEL COMPARISON ON MIMIC-III

Model	Metrics				Settings	
	Jaccard	PRAUC	F1	DDI_rate	DDI?	γ_{ddi}
MLP	0.4871	0.7386	0.6444	0.0808	×	
GRU	0.5019	0.7577	0.6582	0.0809	×	
RETAIN	0.5137	0.7683	0.6766	0.0833	×	
SARMR	0.5039	0.7688	0.6608	0.0833	×	
DRMP(-GCN)	0.5296	0.7754	0.6841	0.0861	×	
DRMP(-GraphSage)	0.5305	0.7742	0.6849	0.0860	×	
DRMP(-Bi)	0.5310	0.7754	**0.6862**	0.0851	×	
DRMP(-GRU)	**0.5312**	**0.7757**	0.6854	0.0865	×	
LEAP	0.4521	0.6749	0.6138	0.0719	✓	
GAMENet	0.5131	0.7627	0.6671	0.0894	✓	0
	0.5172	0.7625	0.6718	0.0883	✓	0.01
	0.5133	0.7628	0.6680	0.0884	✓	0.1
	0.5026	0.7586	0.6767	0.0864	✓	1
SafeDrug	0.5250	0.7691	0.6794	0.0722	✓	$\gamma_{ddi}(0.08)$
	0.5228	0.7634	0.6776	0.0672	✓	$\gamma_{ddi}(0.07)$
	0.5213	0.7647	0.6768	0.0589	✓	$\gamma_{ddi}(0.06)$
	0.5164	0.7604	0.6720	0.0479	✓	$\gamma_{ddi}(0.05)$
	0.5083	0.7556	0.6651	0.0381	✓	$\gamma_{ddi}(0.04)$
DRMP_ddi(-GRU)	0.5323	0.7756	0.6863	0.0775	✓	0.5
	0.5291	0.7736	0.6837	0.0717	✓	1.0
	0.5272	0.7722	0.6819	0.0716	✓	1.2
	0.5244	0.7695	0.6795	0.0651	✓	1.5
	0.5218	0.7699	0.6772	0.0585	✓	2.0
	0.5126	0.7629	0.6693	0.0444	✓	4.0

introduced additional DDI knowledge. Therefore, Table II showing the comparison results is divided into two groups. The first half is the model without additional DDI knowledge, and the second half is the model with additional DDI knowledge. In addition, the models proposed in this paper are denoted as follows according to the composition of different modules. **DRMP**: the proposed drug recommendation model including message propagation module and visit-level attention. **DRMP_ddi**: the proposed drug recommendation model that introduces DDI knowledge and deploys DDI gating mechanism based on DRMP. **DRMP_ddi**$\star$: the proposed drug recommendation model that extends DRMP_ddi to BNN to realize the multiple recommendation and confidence estimation.

C. Metrics

Six evaluation metrics were used to evaluate our model, including Jaccard similarity, PRAUC, F1 score, accuracy, recall and DDI rate. Jaccard similarity is used to measure the proportion of "co-occurring" elements in two sets. It is one of the most commonly used evaluation metrics in the drug recommendation task. F1 score takes into account both accuracy and recall. It can adapt to the possible label imbalance. PRAUC measures the area under the precision-recall curve, which is more convincing.

In addition, precision, recall and DDI rate are also shown. The calculation of DDI rate is as follows:

$$ddi\_rate(\bar{y}) = \frac{\sum_{\forall \bar{y}_i=1} \sum_{\forall \bar{y}_j=1 \& j \neq i} A_{i,j}^{ddi}}{|\{i|\bar{y}_i = 1\}| * (|\{i|\bar{y}_i = 1\}| - 1)} \quad (30)$$

where $|\{i|\bar{y}_i = 1\}|$ is the size of the recommended drug combination.

D. Experimental Results

We find that when we try to reduce the DDI rate in the recommended drug combination, the model performance (evaluated by Jaccard similarity, PRAUC and F1) will decline. This is because there is a large number of DDI phenomenon in the practice data set. Restricting the DDI rate by force will result in the recommendation result not matching the true labels perfectly. Based on the above observation, we divided the comparison models into two groups. The first half of Table II is models without additional DDI knowledge, and the second half is models with additional DDI knowledge. For each model in the second half of the table, multiple experimental results are given for comprehensive comparison. Model settings are given on the right side of the table. *DDI?* indicates whether additional DDI knowledge is introduced into the model. γ_{ddi} is the weight parameter of DDI loss in the loss function, as shown in formula 29. In SafeDrug [5], γ_{ddi} is calculated by a function, which will not be repeated in this paper.

Table II shows the experimental results of models on the data set MIMIC-III. From the comparison of the models in the first half of Table II, it can be seen that our proposed DRMP is superior to the existing models in the main metrics (i.e. Jaccard similarity, PRAUC and F1 score). In the second half of Table II, the performance of models using external DDI knowledge under different DDI loss weights is studied in detail. The following conclusion can be obtained:

- With the increase of γ_{ddi}, the overall trend of the model performance is gradually declining, but the DDI rate in the recommended results is suppressed.
- Compared with GAMENet, the proposed model has higher performance and a lower DDI_rate. This proves that the proposed gating mechanism for DDI is concise and effective.

The performance of the model proposed in this paper is better than the existing baseline. This is due to the model's attention to important visits, important items, and important associations. In addition, the RETAIN model has a relatively strong recommendation ability, which shows that its visit-level and feature-level attention play an important role. Compared with the GAMENet model, SafeDrug has a more significant inhibitory effect on the DDI rate. This may be because GAMENet uses an implicit DDI knowledge embedding method, while SafeDrug uses an explicit DDI loss, which is more conducive to the reduction of the drug DDI rate.

E. Extended to BNN: Multiple Recommendations

The proposed model is extended to BNN to achieve the purpose of multiple recommendations and confidence evaluation. For each sample, the model will give a list of drug combinations and the probability of each drug combination. The sum of the probabilities of all drug combinations is 1. This probability can be regarded as the model's confidence in the recommended

TABLE III
CASE ANALYSIS OF MULTIPLE RECOMMENDATIONS. $\#\bar{y}$, CONFI@K, AC@K AND DDI-INTENSITY@K

Sample	$\#\bar{y}$	Confi@K					AC@K	DDI-intensity@K (×1000)				
1	41	0.19	0.09	0.09	0.08	0.08	0.53	1.39	1.38	1.39	1.12	1.38
2	12	0.59	0.14	0.09	0.05	0.05	0.91	0.14	0.45	0.45	0.14	0.14
3	90	0.07	0.07	0.05	0.05	0.05	0.30	3.36	3.39	3.36	3.43	3.43
4	17	0.43	0.27	0.05	0.05	0.03	0.83	1.58	1.58	1.76	1.76	1.58
5	33	0.19	0.19	0.15	0.06	0.05	0.64	0.91	0.91	0.58	0.58	0.90
6	72	0.17	0.07	0.05	0.04	0.03	0.37	1.83	1.39	1.85	1.83	1.56

results. We rank the recommended drug combinations from high confidence to low confidence.

The model can provide 5 kinds of information for each sample.

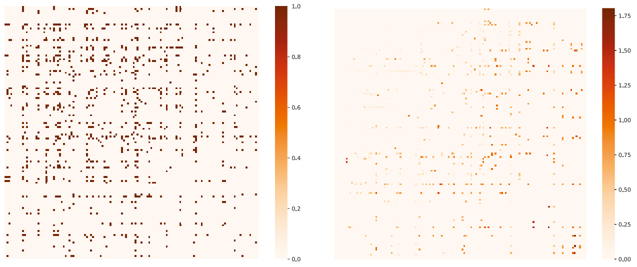
- $\#\bar{y}$: The number of recommended drug combinations.
- **R-drugs[k]**: **R**ecommended **drugs** with the highest K confidence for each sample. $k = 1, 2, \ldots, K$.
- **Confi@K**: The **Confi**dence of drug combinations with the highest K confidence.
- **AC@K**: The **A**ccumulated **C**onfidence of drug combinations with the highest K confidence.
- **DDI-intensity@K**: **DDI intensity** of drug combinations with the highest K confidence. DDI-intensity is calculated in a similar way to DDI_rate, only replacing A^{ddi} with W^{DDI}.

$$\text{DDI-intensity}(\bar{y}) = \frac{\sum_{\forall \bar{y}_i=1} \sum_{\forall \bar{y}_j=1 \& j\neq i} W^{DDI}_{i,j}}{|\{i|\bar{y}_i=1\}| * (|\{i|\bar{y}_i=1\}|-1)} \quad (31)$$

In this paper, let $K = 5$.

4 kinds of information in addition to R-drugs[k] are presented in Table III. It can be seen from Table III that the model gives 12 recommended results for the 2nd sample, and the confidence value of the drug combination with the highest confidence is 0.59 (high confidence). The model gives 90 recommended results for the 3 rd sample, and the confidence value of the drug combination with the highest confidence is only 0.07 (lack of confidence). It can be seen that the more the recommended results, the greater the variance of the probability density function of the solution space, and the lower the confidence of a single solution. Finally, when doctors decide whether to rely on the recommended results of the model, they can make decisions based on a variety of information, including the recommended drug combination itself, the confidence of the model, and the DDI intensity of the drug combination.

Table IV shows R-drugs[k] of a sample. In Table IV, first the diagnosis and ground truth of the sample are given. Subsequently, the first 5 recommended results with the greatest confidence are displayed. The drugs are given in the form of level-3 ATC code. In addition, the following information is shown in the table. The size of the recommended drug combination is denoted as drug#. The number of hits (denoted as hit#) indicates the number of correct drugs in the recommended drug combination. The missing number (denoted as miss#) represents the number of drugs included in the ground truth but not included in the recommended combination. The mistake number (denoted as mistake#) represents the number of drugs included in the recommended set but not in the ground truth. The Jaccard score (denoted as Jaccard) indicates the Jaccard similarity between the recommended set and the ground truth. It is worth noting that although R-drugs[1] has the largest Jaccard score in the shown case, this is not always the case. Multiple recommendations can provide doctors with more choices.



(a) A^{ddi} (b) DDI-weight

Fig. 4. Binary DDI knowledge (i.e., A^{ddi}) and DDI weight matrix.

F. Extended to BNN: Confidence Study

Table V shows the relationship between the recommended results and the confidence threshold (denoted as cft). After the model is extended to BNN, multiple results will be given for each sample. When calculating the metrics, only the recommended result with the greatest confidence is used for each sample. If the confidence of the recommended result is less than the confidence threshold, the sample is discarded because the model's recommendation of this sample is considered unreliable. From Table V, it can be seen that with the increase of the confidence threshold, the recommendation performance (i.e., Jaccard similarity, PRAUC, F1 score) has been improved. This shows that recommendation confidence can help users know whether the model's recommendation results are reliable.

G. DDI Intensity Study

Since the harm of DDI can be divided into high and low, the proposed DDI gating mechanism strives to capture the interaction intensity between different drug pairs. Fig. 4(a) is the DDI external knowledge matrix A^{ddi}, where $A^{ddi}_{i,j} \in \{0, 1\}$, indicating whether there is adverse interaction between the two drugs. Fig. 4(b) is the parameter W^{DDI} in the gating mechanism, $W^{DDI}_{i,j} \geqslant 0$, which indicates the inhibition of drug i to drug j. The drugs are sorted by frequency in Fig. 4. The frequency of drugs on the left/upper side was higher than that on the right/lower side. Comparing Fig. 4(b) with Fig. 4(a), we can see that the DDI intensity in the upper left corner of Fig. 4(b) is generally small. This may be caused by the frequency of drugs. In addition, the values of $W^{DDI}_{i,j}$ and $W^{DDI}_{j,i}$ may not be equal. The occurrence of this phenomenon may be affected by the nature of the drug itself and the frequency of drug occurrence.

The last column of Table III shows the DDI intensity of the recommended results. The DDI intensity of recommendation results is a relative value, which is used to compare the relative intensity of DDI in different recommendation results. It is worth noting that the DDI intensity is not simply based on binary DDI knowledge, but based on the trainable matrix W^{DDI} in the

TABLE IV
CASE ANALYSIS OF MULTIPLE RECOMMENDATIONS

Diagnoses	Unspecified septicemia; Diabetes mellitus without mention of complication, type II or unspecified type, not stated as uncontrolled; Alkalosis; Anemia, unspecified; Other persistent mental disorders due to conditions classified elsewhere; Unspecified essential hypertension; Late effects of cerebrovascular disease, aphasia; Late effects of cerebrovascular disease, hemiplegia affecting unspecified side; Pulmonary collapse; Acute kidney failure, unspecified; Septic shock; Severe sepsis; Gastrostomy status					
Ground truth	A01A A06A B05C A12C C01C A07A C07A A12B J01M B01A C10A C09A J01D D06B N04B J01C A11D (17 drugs)					
R-drugs[k]	Drugs in level-3 ATC code	drug#	hit#	miss#	mistake#	Jaccard
R-drugs[1]	N02B A01A A02B A06A B05C A12A A12C C01C A07A C07A A12B J01M B01A C10A J01D N03A J01C	17	13	4	4	0.6190
R-drugs[2]	N02B A01A A02B A06A B05C A12A A12C C01C A07A C07A A12B B01A C10A J01D N03A J01C	16	12	5	4	0.5714
R-drugs[3]	N02B A01A A02B A06A B05C A12A A12C C01C A07A C07A A12B J01M B01A C10A J01D N03A R03A J01C	18	13	4	5	0.5909
R-drugs[4]	N02B A01A A02B A06A B05C A12A A12C C01C A07A C07A A12B J01M B01A J01D N03A J01C	16	12	5	4	0.5714
R-drugs[5]	N02B A01A A02B A06A B05C A12A A12C A07A C07A A12B J01M B01A C10A J01D N03A J01C	16	12	5	4	0.5714

R-Drugs[k]

TABLE V
ROLE OF RECOMMENDATION CONFIDENCE

cft	Jaccard	PRAUC	F1
0	0.5207	0.7758	0.6761
0.1	0.5231	0.7774	0.6782
0.2	0.5299	0.7807	0.6840
0.3	0.5406	0.7892	0.6932
0.4	0.5491	0.7971	0.7006
0.5	0.5538	0.7985	0.7053

model. W^{DDI} reflects the model's cognition of the importance of these interactions. Compared with the DDI rate of the recommended results obtained based on the binary knowledge, the DDI intensity obtained based on the DDI weight has more reference value when considering drug administration to patients.

VI. CONCLUSION

Drug recommendation is one of the most promising tasks in data mining in the medical field, which has been widely concerned by researchers. Although the existing deep learning-based models have achieved remarkable performance, these models still have some problems, such as low recommendation accuracy, single recommendation result, lack of confidence evaluation, and so on. Therefore, the DRMP model for the drug recommendation task is proposed in this paper. Then, the external DDI knowledge is introduced and a gating mechanism for DDI is proposed to reduce the DDI rate in recommendation results. Finally, the model is extended to BNN, so that the model can make multiple recommendations for patients and provide the confidence of each recommendation result. This will help doctors understand the confidence of the model. Experimental results show that the proposed model not only outperforms the existing models but also provides more sufficient information to help doctors make decisions.

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